

Project Design Phase

Problem – Solution Fit Template

Date	20 February 2026
Team ID	LTVIP2026TMIDS88068
Project Name	Online Payments Fraud Detection using Machine learning
Maximum Marks	2 Marks

Problem – Solution

The Problem–Solution Fit identifies whether the proposed solution effectively addresses the customer’s problem. In the current digital payment environment, users face increasing risks of fraudulent transactions, unauthorized payments, and financial loss.

Customers require secure and reliable online payment systems that can detect fraud instantly without affecting transaction speed. Traditional rule-based systems fail to identify evolving fraud patterns and require manual monitoring.

The proposed solution, an Online Payment Fraud Detection System using Machine Learning, analyzes transaction behavior and automatically predicts fraudulent activities in real time. The system learns from historical transaction data and improves detection accuracy continuously.

By applying machine learning techniques, the solution directly addresses customer concerns related to security, trust, and fast transaction processing. This ensures improved fraud prevention, reduced financial risk, and enhanced confidence in digital payment platforms.

Purpose:

- To solve online payment fraud problems faced by customers and banking institutions.
- To provide an intelligent system that detects fraudulent transactions automatically.
- To improve transaction security and customer trust.
- To enable faster fraud detection using machine learning algorithms.
- To enhance system reliability and real-time monitoring capabilities.

Template:

